

UKRAINE

CONFLICT

VISUALIZATION

Process Book

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PROJECT DEFINITION

We started out wanting to do something related to current events — journalism, geopolitics, ongoing conflicts. Diplomatic alliances came up early, but the data was not there. The war in Ukraine, on the other hand, felt like a natural fit: it is one of the most documented conflicts in recent history, and after some research we found two excellent, openly available datasets that gave us everything we needed.

The idea we settled on was a website for the **informed general public** — people who follow international news and want to understand the conflict beyond the daily headlines. We wanted visitors to be able to explore how the battlefield has shifted month by month, see which countries are supporting Ukraine and by how much, and get a sense of what the war means at the level of individual Ukrainian regions. In short, a single place where users can look at the conflict from three angles: the fighting on the ground, the international response, and the human cost.

Everything on the site flows through a central interactive map, from which users can navigate to per-oblast or per-country detail pages, with a separate page for aggregate statistics and broader trends.

DATA

The two main datasets we used are **ACLED** (Armed Conflict Location & Event Data Project) and the **Ukraine Support Tracker** (UST), maintained by the Kiel Institute for the World Economy. Both are widely considered gold standards in their respective fields and are regularly cited by major international newspapers. We also looked at other datasets, like UCDP and the Ukraine War and Sanctions dataset, but they did not fit well with the direction we were giving our project, so we ended up not using them.

For geographic rendering we used:

- `datasets/geo-countries` — world country boundaries (GeoJSON)
- `wmgeolab/geoBoundaries` — Ukraine ADM1 oblast boundaries (GeoJSON)

Data Preprocessing

Most of the preprocessing was written in Python. A smaller part is in R, inside the EDA script: there we consolidated ACLED's granular actor descriptions into two categories (Russian and Ukrainian military forces) when applicable and standardized date formats. Everything else was handled in Python: downloading and parsing the GeoJSON boundary files, aggregating ACLED events and fatalities by oblast, aggregating UST commitments by donor country, building the event timeline, merging the various European institutional donors listed separately in UST under a single EU entry, and converting all financial figures to EUR using the average exchange rate for each currency over the conflict period.

Running `preprocess.py` outputs a self-contained folder of JSON and GeoJSON files that the website loads directly, with no further processing needed at runtime.

Exploratory Data Analysis

We did the EDA in R, which is fast for working with dataframes and producing plots. Somewhat unusually, EDA came before formal preprocessing in our workflow: rather than exploring already-clean data, we used it as a first pass on the raw datasets to understand what was in them, what questions they could answer, and what preprocessing would actually be worth doing. The plots from EDA.R — event intensity by oblast, the rise of drone strikes over time, the distribution of international aid — shaped what we ended up focusing on in the website.

WEBSITE STRUCTURE

By the time we finished EDA, the shape of the website was already quite clear: a home page with an interactive map and a war timeline, plus the ability to go deeper into individual oblasts or donor countries. We built an initial skeleton with three HTML pages — home, oblast detail, and country detail — plus a shared CSS file.

As we planned out the content, we realized the home page would get too cluttered if we added stats and charts alongside the map. We wanted it to be simple and immediate, so we created a fourth page — Conflict Overview — to hold the aggregate statistics and broader plots. The sections below describe how each page evolved and what challenges came up along the way.

HOME PAGE

The home page is built around a single interactive map and a war timeline. We always wanted one map rather than several — it is simpler to navigate and gives the user a unified picture of the conflict. The main challenge was fitting three distinct interaction modes onto that map: country-level data, Ukrainian oblasts, and the battlefield.

Early prototype. Our first idea was to use zoom level as a way to switch between views: zoom out to see the world map with clickable countries, zoom in to see Ukraine's oblasts. While elegant in concept, in practice it was awkward to navigate, and more importantly it made it impossible to show battlefield events without the map becoming visually cluttered. We replaced it with a mode-switcher, with an adapting zoom and letting the user choose which layer to activate.

Battlefront and event display. The zooming was adapted to the nature of the conflict: the frontline has only moved a few tens of kilometers over three years, so a single zoom level works well for every month of the war. We display events at monthly rather than daily granularity, since day-to-day battlefield changes are nearly imperceptible — monthly aggregates are where the patterns become visible. A *play* button lets users animate forward from any selected month, which helps convey the spatial progression of the war without requiring the map to be linked directly to the timeline, which was an initial idea.

Inferring the frontline. We had originally planned to overlay published frontline polygons, but finding temporally consistent data at the right granularity proved harder than expected. We then tried to compute the front from battle event positions — and quickly realized we already had what we needed: filtering ACLED events to just “Battles” produces a near-continuous strip of points that closely matches published front maps. That observation led us to add an event-type filter on the map, so users can isolate battles, explosions, drone strikes, or other categories independently. A density heatmap visualization complements the point scatter — sometimes a single impression of intensity is more useful than individual event dots.

Month comparison. Since the front moves slowly, differences between two months can be subtle. We added a comparison mode where the user selects two months and overlays them: events from the second month are circled in white, making the shift legible at a glance (Figure 1).

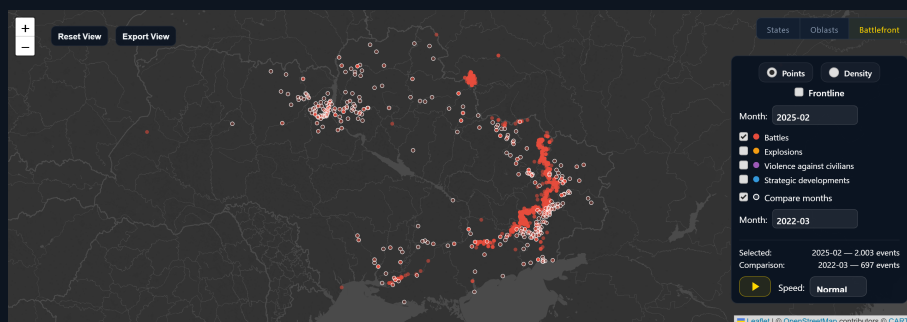


Figure 1: Comparing an early-2022 month with a 2025 month reveals the full arc of the conflict: Russian forces no longer threatening Kyiv from the north, while the eastern front has gradually shifted in Russia’s favor.

Since the map has a lot of configurable state — mode, month, event type, density toggle — we also added a snapshot export button so users can save and share any configuration they find informative.

CONFLICT OVERVIEW

This page brings together the aggregate statistics and charts from ACLED and UST. We originally wanted to include a section on civilian casualties, but we ran into an issue with how ACLED categorizes events. The “Violence against civilians” category covers cases where the perpetrator is a non-state actor, a civilian group, or a police force — not conventional military operations. Civilian deaths from drone strikes, for example, fall under “Explosions/Remote violence.” The categories do not map cleanly to the question we wanted to answer, so we ended up giving that dimension much less space.

For international aid, we built on the sketch from Milestone 2 — a breakdown by donor country — and extended it with a cumulative time series and a GDP-normalized ranking (Figure 2). A key challenge was the large disparity in aid volumes between the United States and other donors. To keep the cumulative time series readable, we made the y-axis scale adaptive to the selected countries. The GDP normalization required sourcing additional data not originally planned for, but proved worthwhile: the relative ranking differs substantially from the absolute one, and that contrast is an insight worth surfacing.

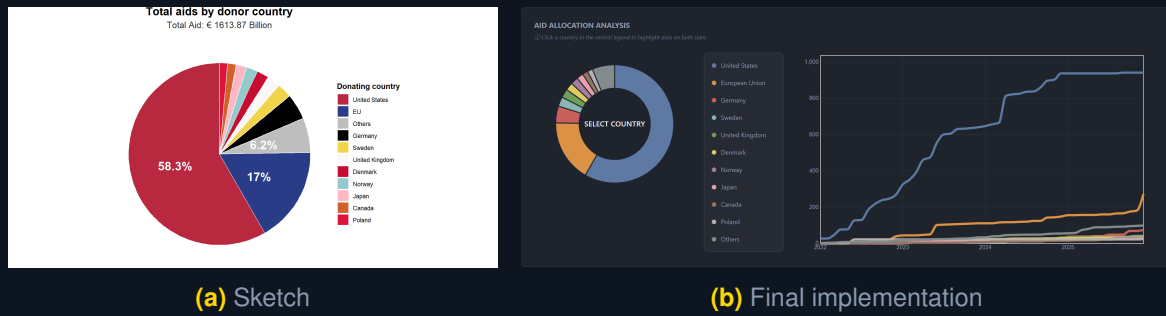


Figure 2: Aid distribution across donor countries.

COUNTRY PAGE

The country page shows a per-donor dashboard based on the UST data, closely following the sketch from Milestone 2. Of the four pages, this was the most straightforward to implement: the data only needed basic aggregation, and the charts — donut, time series, summary statistics — follow standard D3 patterns. We kept the hover interaction on the donut chart (highlight a segment, see the exact value) consistent across the whole website.

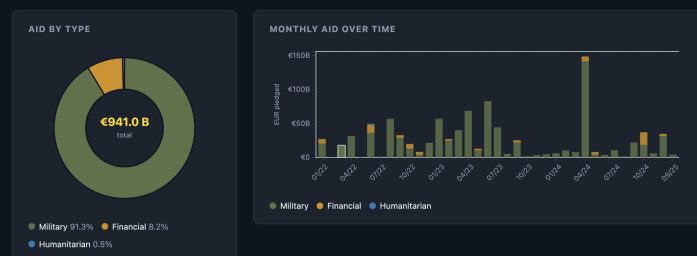


Figure 3: Partial screenshot of the country page for the United States, showing the aid breakdown by category and its evolution over time.

OBLAST PAGE

The oblast page is a dashboard for the conflict events recorded in a given region: headline numbers, a breakdown by event type, and a monthly time series. The time series is the

dynamic counterpart of the plot we had already produced in R during EDA — Figure 4 shows the prototype alongside the final result for Kyiv.



Figure 4: R prototype (left) vs. final website plot (right) for the Kyiv event timeline.

Fatality pictograms. The most challenging element of the page was the fatality pictogram. We wanted human-figure icons to give deaths more visual weight than a bare number, but oblasts vary enormously in scale — some have single-digit fatalities, others have tens of thousands — so a uniform icon scale does not work across the board.

We explored several approaches and ultimately abandoned cross-oblast comparability in favor of per-oblast legibility. Each region uses a scale unit (one figure = 100, 500, 2 000 or 5 000 deaths, depending on the oblast) chosen to produce a readable grid for that specific region. The scale unit is shown explicitly, and different units are size-coded, so even without reading the legend users can tell that pictograms across oblasts are not directly comparable.

PEER ASSESSMENT

Daniele Giuli

Daniele handled data preprocessing and exploratory data analysis (`preprocess.py` and `EDA.R`). He also designed the overall website structure and visual style, including the four HTML pages, the CSS stylesheet, and the core of the homepage featuring the timeline and map. He has built most of the Countries page and contributed partially to the Oblasts page. He also wrote the process book.

Ali Shenaskhosh

Ali produced some of the sketches for the Milestone 2 submission and managed the GitHub repository and overall project structure. He was solely responsible for the Battlefield visualization, developing all of its components. He recorded and scripted the screencast, and carried out final refinements across the website.

Chaewon Yoon

Chaewon completed the Countries page started by Daniele and took primary responsibility for the Oblasts page. She implemented the Conflict Overview page from scratch to end where she developed the entire suite of user interactive functions. She also wrote the final README.